Adaboost Classification

x_1	x_2	y	w
1	2	+1	0.25
2	1	-1	"
4	4	+1	"
5	3	-1	"

1) Initialize sample weights

$$w^{(i)} = \frac{1}{N} \text{ for all samples}$$

$$= \frac{1}{4} = 0.25$$

Possible splits for x_1 → 1.5, 3, 4.5

for →

$x_1 < 1.5$

N ↙ ↘ Y

S_2, S_3, S_4 $\boxed{-1}$ $\boxed{+1}$ S_1

$Error^{(1)} = 0$

$\epsilon^{(2)} = 0$

$\epsilon^{(3)} = w^{(3)} = 0.25$

$\epsilon^{(4)} = 0$

from this

$$\boxed{f_t = \underset{t}{\operatorname{argmin}} \sum_i w^{(i)} I(y^{(i)} \neq f(x^{(i)}; \theta_t))}$$

$f_t = 0.25$

for split = 3

$\boxed{x_1 < 3}$

N Y

S_3, S_4 $\boxed{+1}$ $\boxed{-1}$ S_1, S_2

Total error = 0.5

for split = 4.5

$x_1 < 4.5$

N Y

S_4 $\boxed{-1}$ $\boxed{+1}$ S_1, S_2, S_3

not below 2 + there are 2 + one will be + another

total error = 0.25

for $x_2 \rightarrow 1.5, 2.5, 3.5$

1.5

$x_2 < 1.5$

error = 0.75

S_1, S_3, S_4 $\boxed{-1}$ $\boxed{+1}$ S_2

$x_2 < 2.5$

error = 0.5

S_3, S_4 $\boxed{-1}$ $\boxed{+1}$ S_1, S_2

$x_2 < 3.5$

error = 0.75

S_3 $\boxed{+1}$ $\boxed{-1}$ S_1, S_2, S_4

$$\lambda_1 = \frac{1}{2} \ln\left(\frac{1-\varepsilon}{\varepsilon}\right) = \frac{1}{2} \ln\left(\frac{1-0.25}{0.25}\right)$$

$$= \frac{1}{2}\ln(3) = 0.5493$$

$$w^{(i)} = 0.25 \times e^{-0.5493}$$
$$= 0.144$$

$$w^{(i)} = 0.25 \times e^{0.5493}$$
$$= 0.433$$